

Yifu Ding

✉ eveedyf@gmail.com 📞 +86 18868001866 (🇨🇳) 🌐 Homepage 🎓 Google Scholar

Ph.D. candidate working on AI Infra, including model compression, inference optimization, and high-performance kernels, with a focus on algorithm-system co-optimization for low-bit quantization, pruning, knowledge distillation, KV Cache optimization, and high-performance kernel development (CUDA/Triton). Published **36** papers in international journals and conferences, including ICML, ACL, NeurIPS, ICLR, CVPR, and TPAMI, with **9** as first, co-first, or corresponding author; **1.6K+ citations**, h-index 20, and 2 granted national patents.



Education

Nanyang Technological University, Singapore. Visiting Ph.D.	11/2024 – 11/2026
Beihang University, China. Computer Science and Technology, Ph.D.	09/2021 – 12/2026
Beihang University, China. Computer Science and Technology, B.Eng. GPA: 3.80/4.0	09/2017 – 07/2021

Publication Highlights

Efficient Training and Inference for Foundation Models

- Attribution-Guided and Coverage-Maximized Pruning for Structural MoE Compression. **Y. Ding**, J. Wang, G. Yang, Y. Jing, J. Guo, X. Liu, D. Tao. **ICML 2026**, *Spotlight*. [PDF] [Code]
 - ▶ Propose joint pruning and quantization framework for deployment-friendly **Mixture-of-Experts (MoE)s**;
 - ▶ **94.7%** original accuracy on QA, math, and code benchmarks; **20×** throughput and **5.27×** storage reduction with 4-bit quantization.
- DA-KD: Difficulty-Aware Knowledge Distillation for Efficient Large Language Models. C. He, **Y. Ding**, J. Guo, R. Gong, H. Qin, X. Liu. **ICML 2025**. [PDF]
 - ▶ Efficient SFT distillation, propose stratified difficulty-aware distillation with a bidirectional discrepancy loss;
 - ▶ Distills Qwen2.5-7B to 1.5B on Dolly-15K with **50%** less training time and near-teacher performance.
- Singular Proxies for Adaptive Caching in Diffusion Language Models. W. Sun, R. Tu, **Y. Ding**, Z. Jin, J. Liao, S. Liu, D. Tao. **ICML 2026**. [PDF]
 - ▶ To enable KV cache reuse for diffusion models, propose hierarchical adaptive caching mechanism;
 - ▶ **8×** throughput on math/code benchmarks with LLaDA-8B and Dream-7B; **28×** with parallel decoding.
- A Survey of Low-Bit Large Language Models: Basics, Systems, and Algorithms. R. Gong, **Y. Ding**, Z. Wang, C. Lv, X. Zheng, et al. **Neural Networks 2025**. [PDF]
 - ▶ Comprehensive low-bit LLM survey covering **23** inference frameworks, **5** toolkits and benchmarks, systematically introducing quantization fundamentals, systems, training, and inference algorithms and techniques.
- LLMCBench: Benchmarking Large Language Model Compression for Efficient Deployment. G. Yang, C. He, J. Guo, J. Wu, **Y. Ding**, A. Liu, H. Qin, P. Ji, X. Liu. **NeurIPS 2024**, *Spotlight*. [PDF] [Code]
 - ▶ Evaluates **7** compression methods, **18** LLM architectures, **11** datasets, and **3** deployment platforms.

High-Performance Kernels and Hardware-Aware Acceleration

- Diagonal-Tiled Mixed-Precision Attention for Efficient Low-Bit MXFP Inference. **Y. Ding**, X. Zhang. **EDGE @ CVPR 2026**. [PDF] [Code]
 - ▶ Triton-based Micro-scaling Floating-Point (MXFP) mixed-precision attention kernel;
 - ▶ Substantial acceleration with lossless LongBench generation quality on NVIDIA B200.
- Accurate and Efficient Binarized Transformer by Algorithm-Hardware Co-design. **Y. Ding**, X. Liu*, S. Jin, J. Guo, J. Lu. **TPAMI** under review. [PDF] [Code]
 - ▶ First high-performance CUDA kernels for ultra low-bit binary/ternary quantized Transformers;
 - ▶ **16–24×** over cuBLAS FP16 and over **80,000 tokens/s** prefill throughput on 2B LLMs.
- Low-bit FlashAttention Accelerated Operator Design Based on Triton. J. Du, J. Guo, **Y. Ding**. **ECLR@ICCV 2025**. [PDF]
 - ▶ Triton-based mixed-precision FlashAttention integrating quantization, dequantization, and attention;
 - ▶ **2.4×** kernel and **1.2×** end-to-end speedup over FlashAttention2 with preserved accuracy.
- QuS: Towards High-Performance EfficientViT on FPGA by Quantization and Streamline Co-Design. Y. Zeng, **Y. Ding**, J. Guo, H. Qin, Y. Xu, N. Li, Y. Guo, X. Liu. [PDF]
 - ▶ For extremely acceleration of ViT on FPGA, propose DSP packing, low-buffer streamline, and distribution-aware quantization; ▶ Over **2,200 FPS**, **3.6×** over Jetson AGX Orin, and up to **24%** higher 4-bit accuracy.

Inference Acceleration

10. MrFlow: Multi-Resolution Flow Matching: Training-Free Diffusion Acceleration via Staged Sampling. X. Zheng, X. Liu, **Y. Ding**, W. Feng, J. Lin, J. Guo, H. Qin. *arXiv preprint*, 2026. [PDF] [Code] [HF]
 - ▶ Propose training-free multi-resolution staged sampling for flow-matching text-to-image diffusion;
 - ▶ **10.3×** e2e speedup on Qwen-Image, **8.25×** on FLUX.1-dev, and **25×** when combined with Pi-Flow.
11. Dynamic Parallel Tree Search for Efficient LLM Reasoning. **Y. Ding**, W. Jiang, S. Liu, Y. Jing, J. Guo, Y. Wang, J. Zhang, Z. Wang, Z. Liu, B. Du, X. Liu, D. Tao. **ACL (Main)** 2025. [PDF] [Code]
 - ▶ To speed up LLM reasoning, propose parallel Tree-of-Thoughts framework with tree-structured KV caching;
 - ▶ **2–4×** **speedup** on math and code while matching sequential MCTS accuracy.

Multimodal and Video Generation

12. QVGen: Pushing the Limit of Quantized Video Generative Models. Y. Huang, R. Gong, J. Liu, **Y. Ding**, C. Lv, H. Qin, J. Zhang. **ICLR** 2026. [PDF] [Code]
 - ▶ Propose quantization-error mitigation for higher-quality and stable-training video diffusion QAT;
 - ▶ First 4-bit models matching full-precision quality across four mainstream video diffusion models.
13. VORTA: Efficient Video Diffusion via Routing Sparse Attention. W. Sun, R. Tu, **Y. Ding**, J. Liao, Z. Jin, S. Liu, D. Tao. **NeurIPS** 2025. [PDF] [Code]
 - ▶ Propose dynamic routing attention with sparse long-context capturing to accelerate video diffusion;
 - ▶ **1.76×** **lossless speedup** on VBench and up to **14.4×** with caching and step distillation.
14. MoDES: Accelerating Mixture-of-Experts Multimodal LLMs via Dynamic Expert Skipping. Y. Huang, Z. Wang, Z. Yuan, **Y. Ding**, R. Gong, J. Guo, X. Liu, J. Zhang. **CVPR** 2026. [PDF] [Code]
 - ▶ Propose training-free expert skipping driven by modality heterogeneity;
 - ▶ On Qwen3-VL, skipping 88% tokens gains **10.67%** accuracy; **2.16×** prefill and **1.26×** decoding speedup.
15. QuantSR: Accurate Low-Bit Quantization for Efficient Image Super-Resolution. H. Qin, Y. Zhang, **Y. Ding**, Y. Liu, X. Liu, M. Danelljan, F. Yu. **NeurIPS** 2023, *Spotlight*. [PDF] [Code]
 - ▶ Deep dynamic quantization with parameter redistribution for efficient image super-resolution;
 - ▶ Outperforms prior methods on 4-bit quantization with flexible accuracy-efficiency trade-offs.

Technical Skills

Model Compression and Acceleration: Extensive experience in low-bit quantization, pruning, knowledge distillation, KV-cache optimization, efficient decoding, and diffusion model acceleration. full-stack development experience across algorithm design, kernel implementation, end to end deployment, and performance analysis.

High-Performance Kernels: Strong background in efficient kernel and systems optimization, including mixed-bit and sparse computation, kernel fusion, custom CUDA C++, Triton, and FPGA-oriented implementation. Experienced in profiling GPU workloads, diagnosing memory/computation bottlenecks, and optimizing latency and throughput with Nsight Systems.

Model Training and Deployment: Strong proficiency of modern LLM training and deployment frameworks and toolkits, including PyTorch, Hugging Face Transformers, TRL, PEFT, Unsloth, Accelerate, DeepSpeed, bitsandbytes, and vLLM. Experienced in LoRA/PEFT fine-tuning, RLHF/GRPO-style training workflows, low-bit inference, and deployment under practical memory, latency, and hardware constraints.

Academic Services

Workshop Chairs: Program Chair, *ECLR @ ICCV 2025*; Local Arrangement Chair, *GLOW @ IJCAI 2024*; Publicity Chair, *EMCLR @ ACM MM 2024*.

Student Editor: Co-edited the Springer CCIS volume *Generalizing from Limited Resources in the Open World*, 2024.

Awards and Funding

Outstanding Doctoral Academic Fund , Beihang University. CNY 40,000.	2025
State Scholarship Fund , China Scholarship Council. SGD 26,400 (approx. CNY 140,000).	2024
National Scholarship for Graduate Students , Ministry of Education of the P.R. China. CNY 50,000.	2024
Outstanding Academic Achievement Award , Beihang University.	2023
Doctoral Academic Scholarship , First Prize, Beihang University.	2022